

DNS

BY

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Introduction

The internet is used every day for activities such as browsing websites, watching videos, sending emails, using social media, and making online payments. Although these activities feel simple to us, many processes happen in the background to make them possible. One of the most important of these processes is the Domain Name System (DNS).

DNS makes the internet easier to use by converting human-friendly domain names, such as google.com or amazon.in, into IP addresses that computers use to communicate with servers. Without DNS, users would have to remember numerical IP addresses for every website they wanted to visit.

Understanding DNS is also important in networking and cybersecurity. DNS is involved in almost every connection made on the internet, and attackers can sometimes misuse it for activities such as phishing, malicious redirects, and communication with harmful servers. Therefore, learning how DNS works helps us understand not only how websites are accessed but also how network communication can be monitored and protected.

This report explains the basic concept of DNS, the structure of domain names, common DNS record types, the process of resolving a domain name into an IP address, DNS caching and TTL, and the importance of DNS in cybersecurity.

What is DNS?

DNS stands for Domain Name System. In simple words, DNS is like the phonebook of the internet. Every device connected to the internet has an IP address. An IP address is a number used to identify a device or server on a network. For example: 142.250.183.14. Remembering numbers like this for every website would be difficult. Instead, we use easy-to-remember names such as:

- google.com
- youtube.com
- amazon.in
- wikipedia.org

DNS connects these easy-to-remember domain names to their corresponding IP addresses. For example, when you type www.google.com into your browser, your computer uses DNS to find the IP address of Google's server. Once it finds the address, your browser can connect to the server and load the website.

So, in simple terms: Domain Name → DNS → IP Address → Website

A screenshot of a quiz interface. At the top, it says "Answer the questions below" in green. Below that, the question "What does DNS stand for?" is displayed. There is a dark input field for the answer, and to its right is a green button with a checkmark and the text "Correct Answer".

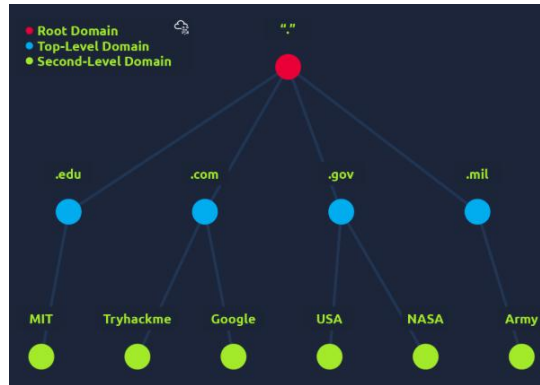
Answer the questions below

What does DNS stand for?

✓ Correct Answer

Domain Hierarchy

A domain name is made up of different parts. Understanding these parts helps us understand how the internet organizes websites. For example, consider: www.amazon.in. This name can be divided into different sections.



1. Top-Level Domain (TLD)

The TLD is the last part of a domain name. In amazon.in, the TLD is: .in

Some common TLDs include:

- .com – commonly used by businesses and general websites
- .org – commonly used by organizations
- .edu – commonly associated with educational institutions
- .gov – used by government organizations
- .in – India
- .uk – United Kingdom
- .ca – Canada

There are also many newer TLDs such as .online, .shop, .tech, and .website. The important thing to remember is that the TLD helps identify the type or geographical association of a domain.

2. Second-Level Domain

The part directly before the TLD is called the Second-Level Domain (SLD). For example: amazon.in

Here:

- amazon → Second-Level Domain
- .in → TLD

The second-level domain is usually the name chosen by the person, company, or organization registering the domain.

3. Subdomain

A subdomain is placed before the main domain name. For example: mail.google.com

Here:

- mail → Subdomain
- google → Second-Level Domain
- .com → TLD

Companies use subdomains to separate different services or sections of their websites.

For example: support.example.com could be used for customer support, while: shop.example.com could be used for an online store. A domain can have multiple levels of subdomains. For example: login.accounts.example.com here, login and accounts are subdomains.

Answer the questions below

What is the maximum length of a subdomain?

✓ Correct Answer 

Which of the following characters cannot be used in a subdomain (3 b _ -)?

✓ Correct Answer

What is the maximum length of a domain name?

✓ Correct Answer

What type of TLD is .co.uk?

✓ Correct Answer

DNS Record Types

DNS does more than simply find the IP address of a website. It stores different types of information using different DNS records. The most common DNS record types are A, AAAA, CNAME, MX, and TXT.

1. A Record

An A record connects a domain name to an IPv4 address.

For example: example.com → 93.184.216.34. When your computer asks for the A record of a website, DNS can provide its IPv4 address. You can think of an A record as: Website name → IPv4 address

2. AAAA Record

An AAAA record works similarly to an A record, but it is used for IPv6 addresses. IPv6 addresses are longer because IPv6 was designed to provide a much larger number of available addresses. For example: example.com → 2606:2800:220:1:248:1893:25c8:1946. So:

- A record → IPv4
- AAAA record → IPv6

3. CNAME Record

A CNAME (Canonical Name) record points one domain name to another domain name. For example, a company might have: www.example.com pointing to: example.com. Instead of storing a separate IP address for both names, the CNAME can tell DNS: "For this domain, look at that other domain." The DNS system can then find the IP address of the other domain.

CNAME records are commonly useful when websites or services are hosted using third-party platforms.

4. MX Record

An MX (Mail Exchange) record is used for email. It tells the internet which mail servers are responsible for receiving emails for a domain. For example, if a company uses: user@company.com the MX records for company.com tell other mail servers where emails for that domain should be delivered.

MX records can also have a priority value. If there are multiple mail servers, the priority helps determine which server should be tried first. This provides backup options if the primary mail server is unavailable.

5. TXT Record

A TXT record stores text information related to a domain. Although it is called a "text" record, it has several important uses. One common use is email security. For example, a domain can use TXT records to tell email providers which servers are allowed to send emails on its behalf. This helps reduce problems such as email spoofing and spam.

TXT records are also commonly used to verify ownership of a domain. For example, when you connect your website to an online service, that service may ask you to add a specific TXT record to prove that you control the domain.

Answer the questions below

What type of record would be used to advise where to send email?

✓ Correct Answer

What type of record handles IPv6 addresses?

✓ Correct Answer

Making a Request

What Happens When You Visit a Website?

Now let's look at what actually happens when you type a website address into your browser. Suppose you type: www.youtube.com. Your computer needs to find the IP address of YouTube's server before it can connect to it. The process generally works like this:

Step 1: Your Computer Checks Its Cache

Your computer first checks whether it already knows the IP address of the website. If it recently visited the website, the answer may already be stored in its DNS cache. If the IP address is available, your computer can use it without making another DNS lookup.

Step 2: Your Computer Contacts a DNS Resolver

If the answer isn't stored locally, your computer sends a DNS request to a DNS resolver. The resolver is usually provided by your internet service provider (ISP), but other public DNS services can also be used. The resolver also has its own cache. For example, if many people recently requested the IP address of YouTube, the resolver may already have the answer stored. If it has the answer, it sends it back immediately. If it doesn't, it needs to search for the answer.

Step 3: The Resolver Contacts a Root DNS Server

If the resolver doesn't know the answer, it starts by contacting a root DNS server. The root server doesn't normally provide the final IP address. Instead, it tells the resolver where to find information about the required TLD. For example, if the request is for: www.youtube.com the root server sees **.com** and directs the resolver toward the DNS servers responsible for the .com domain.

Step 4: The TLD Server Provides the Next Direction

The resolver then contacts the .com TLD server. The TLD server knows which authoritative DNS server is responsible for the requested domain. For example, it can tell the resolver: "The DNS information for youtube.com can be found at these authoritative name servers." The resolver then contacts the appropriate authoritative server.

Step 5: The Authoritative DNS Server Gives the Answer

The authoritative DNS server is the server that stores the official DNS records for a domain. It can provide the requested record. For example: www.example.com → 93.184.216.34. The resolver receives this information and sends the answer back to your computer. Your browser can now connect to the server using the IP address and load the website.

DNS Caching and TTL

DNS uses caching to make the process faster.

Imagine that millions of people request the IP address of Google every minute. It would be inefficient for DNS servers to perform the entire lookup process every time. Instead, DNS servers temporarily store answers they have already found. Each DNS record has a value called TTL (Time To Live). TTL tells the DNS server how long it can keep the information in its cache. For example, if a DNS record has a TTL of: 3600 seconds the record can normally be cached for 1 hour before it needs to be checked again.

Caching helps:

- Reduce DNS traffic
- Make websites load faster
- Reduce the workload on DNS servers
- Improve the overall efficiency of the internet

Answer the questions below

What field specifies how long a DNS record should be cached for?

✓ Correct Answer

What type of DNS Server is usually provided by your ISP?

✓ Correct Answer

What type of server holds all the records for a domain?

✓ Correct Answer

Practical

We are supposed to use the record type commands in the command prompt and answer the value for given question. Here we learn that how the record type commands are used and their function.

Answer the questions below

What is the CNAME of shop.website.thm?

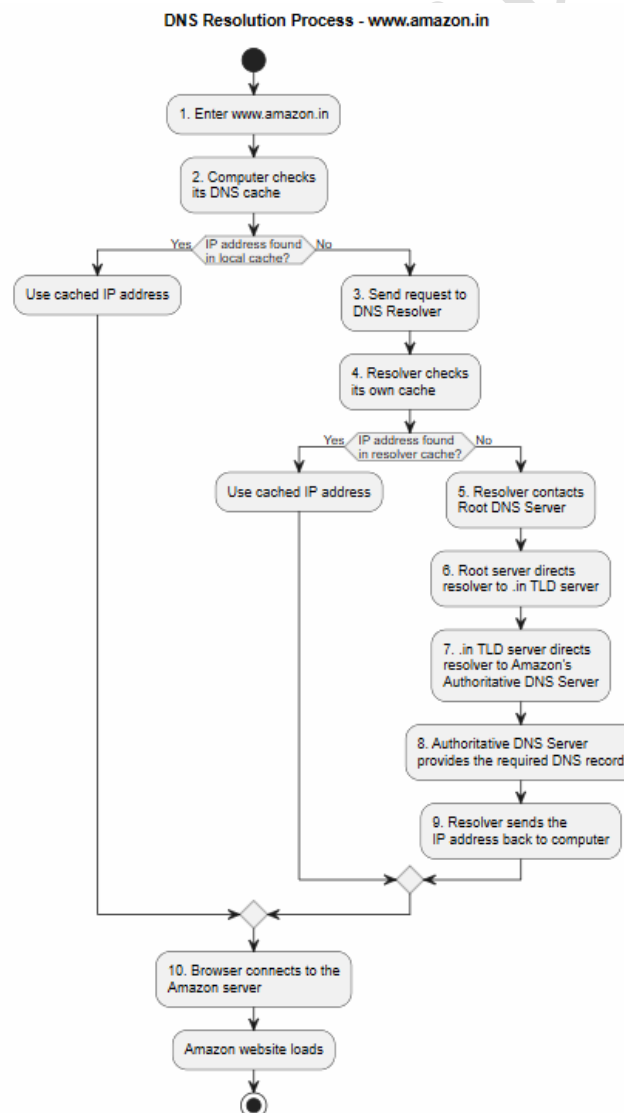
What is the value of the TXT record of website.thm?

What is the numerical priority value for the MX record?

What is the IP address for the A record of www.website.thm?

Simple Example of the Complete Process

Imagine you type: www.amazon.in into your browser. The process can be simplified as:



This entire process usually happens very quickly, often without the user even noticing it.

Why is DNS Important in Cybersecurity?

DNS is an important part of cybersecurity because attackers can also abuse it.

For example, attackers may create fake domains that look similar to legitimate websites.

A fake website might use a domain that looks like: paypa1.com instead of: paypal.com

This technique can be used in phishing attacks to trick users into entering usernames, passwords, or financial information.

Security professionals also analyse DNS information to:

- Identify suspicious domains
- Investigate phishing websites
- Find connections between domains and servers
- Detect potentially malicious infrastructure
- Investigate malware communication
- Identify unusual DNS requests

Therefore, understanding DNS is useful not only for browsing the internet but also for network security, ethical hacking, and cybersecurity investigations.

Conclusion

DNS is one of the fundamental systems that makes the internet easy and practical to use. Instead of remembering complicated IP addresses, users can simply enter familiar domain names such as youtube.com, google.com, or amazon.in, while DNS handles the process of finding the appropriate server.

As discussed in this report, DNS uses a hierarchical system involving DNS resolvers, root servers, TLD servers, and authoritative DNS servers. Different DNS records, such as A, AAAA, CNAME, MX, and TXT, provide different types of information. DNS caching and TTL also help make the process faster and reduce unnecessary network traffic.

From a cybersecurity perspective, DNS is particularly important because it can provide useful information about network activity and can also be abused by attackers. Understanding DNS helps security professionals investigate suspicious domains, identify potential phishing websites, analyze malicious infrastructure, and understand how devices communicate across networks.

Overall, DNS may work silently in the background, but it is an essential part of almost every internet connection. Having a clear understanding of how DNS works provides a strong foundation for learning more advanced topics in computer networking, ethical hacking, and cybersecurity.